# Yingrui Hou

## Data Science & Scientific Data Engineering

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https://yingrui-hou.github.io/PersonalPage/

### Profile

Physics PhD with postdoctoral experience in statistical modelling, CatBoost workflows and learned-model calibration. Learned TensorFlow through adapting inherited code, verifying models and moving computation to a shared GPU workstation. Enjoys diagnosing problems, testing solutions and advancing collaborative projects; seeks to apply these skills to data applications that help improve people's everyday lives.

### Core skills

- Model assessment: deep-learning calibration, image-recognition model validation, real-data performance checks, train/test diagnostics and bias correction.
- Modelling: likelihood and mixture models, statistical inference, uncertainty analysis, CatBoost/gradient boosting, feature selection and distribution reweighting.
- Workflows: inherited-code adaptation, reusable processing, Shell/YAML/CMake, Linux and shared GPU workstation setup/management.
- Programming & computing: Python, NumPy, pandas, scikit-learn, statsmodels; C/C++, ROOT/RooFit; TensorFlow and GPU numerical evaluation.

### Research experience

**Postdoctoral Researcher | 2021–2026**

CNRS — LAPP, Annecy (2021–2023); LPC/LPCA, Clermont-Ferrand (2023.12.01–2026.05.31). Research in LHCb and GRAiNITA collaborations.

- Developed a Shell pipeline for beam-test acquisition, storage, quick calibration and initial quality checks, connecting detector readout, tracking and timing calibration. Participated in on-site setup and data taking; assisted subsequent fitting and measurement-effect corrections.
- Developed CatBoost selection with reweighting, feature pruning, five-fold assessment and train/test KS checks; scanned a measurement-oriented threshold using fitted yields and selection efficiencies.
- Aligned simulated, control and target datasets through gradient-boosting reweighting and calibration; propagated residual modelling differences into uncertainty assessments.
- Diagnosed incorrect ECAL photon truth associations; revised cluster-energy matching, tested multiple simulated decay channels and improved thread safety through review. The merged change supported subsequent Run 3 analyses and reconstruction-performance checks.

**Doctoral research training | 2016–2021**

University of Chinese Academy of Sciences — joint Master and PhD in Particle Physics. Training in quantitative problem formulation, scientific computation, experimental validation and communication of research results.

### Education

- Joint Master and PhD in Particle Physics — University of Chinese Academy of Sciences, 2016–2021.
- BSc in Applied Physics — China University of Mining and Technology, 2010–2014.

## Selected project evidence

### ECAL diagnosis and integrated software correction

Traced anomalous simulated photon resolution to incorrect truth associations under high occupancy. Revised matching using normalised truth-particle energy deposits within the photon cluster. Checked identities, energies and event histories across several simulated decay channels. Improved thread safety following review; the MR was merged and used in Run 3 analyses and reconstruction-performance checks.

### Calibrating learned-model outputs

Validated and calibrated a deep-learning flavour-tagging model using real data for subsequent analysis. Transferable method: check the reliability of learned-model outputs before using them in a new analysis.

### Machine learning under dataset shift

Selected Bs → phi gamma with simulated signal and real-data background. Checked correlations and dataset shifts with control data and hep_ml. Used five folds for feature and hyperparameter selection, with train/test KS diagnostics. Scanned S/sqrt(S+B) using fitted baseline yields and pooled train/test efficiencies; final assessment came from the five-fold results.

### Learning TensorFlow and delivering GPU computation

Learned TensorFlow from inherited code; personally ported amplitude models, added efficiency, resolution and bias effects, and validated calculations against mature C++ code at identical sampling points. Resolved discrepancies by aligning variable types; checked simulated observables against real data with KS tests. Set up a shared GPU workstation in half a day, then achieved a GPU run in another half-day. Contributed to testing a collaborator-developed fitter.

For flavour tagging and ECAL clustering, participated in model-construction and training discussions for downstream calibration and validation; did not train these two models.

## Learning, delivery & collaboration

- Learning by implementation: extended unfamiliar TensorFlow code with new research models and checked it on generated data; applied CatBoost and Geant4 in other projects.
- Delivery: agreed component interfaces and variable definitions before coding; reassessed priorities and timelines with collaborators as data processing exposed new issues.
- Collaboration: GitLab development and mutual merge-request review; identified a fitter sign error through mathematical checks. Weekly planning and rapid technical discussions; research presentations at GDR-InF 2022 and FCC/DRD France 2025.

Proposed first assignment: agree the decision and success criteria, check data quality, establish a baseline and deliver a focused validation study with findings and next steps.

## Languages & mobility

Chinese: native | English: C1 | French: B1–B2. Long-distance cycling and travel; open to relocation in Europe.

## Project references

- ECAL diagnosis and integrated software correction (https://yingrui-hou.github.io/PersonalPage/index.html#work-6)
- Calibrating learned-model outputs (https://yingrui-hou.github.io/PersonalPage/index.html#work-9)
- Machine learning under dataset shift (https://yingrui-hou.github.io/PersonalPage/index.html#work-3)
- Learning TensorFlow and delivering GPU computation (https://yingrui-hou.github.io/PersonalPage/index.html#work-8)